

Table 1: Zero-shot only (do not compare cells to the few-shot companion table). Falcon-Mamba BA-BILong ablation results. Accuracy is exact-match (%) averaged over QA1-QA3 (50 examples per task and length). *No cache, zero-shot:* plain prompt, no [CACHE] (eval_babilongv2, few-shot-examples 0); reproduced from metrics/zero_shot_baseline/falcon_mamba_next_token_qa1_qa3.json. *StateCache* rows: cache_acc from metrics/{run_id}.json (inline scaffold, no few-shot block); layer sweep rows from metrics/layer-experiments/{run_id}.json; L6 follow-up sweeps from metrics/l6-sweeps/{run_id}.json.

Method	0k	1k	2k	4k	8k	16k	Avg.
No cache, zero-shot	44.7	25.3	24.7	16.7	18.7	19.3	24.9
Loss-selected, $k = 1$, L62	22.7	11.3	20.0	16.0	20.0	14.7	17.4
Loss-selected, $k = 4$, L62	16.7	18.0	15.3	12.7	14.7	14.7	15.3
Loss-selected, $k = 8$, L62	20.0	18.7	15.3	21.3	16.7	22.0	19.0
Loss-selected, $k = 16$, L62	18.0	12.7	22.7	11.3	20.0	13.3	16.3
Loss-selected, $k = 32$, L62	23.3	10.0	19.3	12.7	18.7	19.3	17.2
Loss-selected, $k = 64$, L62	15.3	18.7	25.3	18.0	14.0	21.3	18.8
Loss-selected, $k = 4$, L62, train up to 8k	20.0	14.7	16.7	15.3	13.3	15.3	15.9
Loss-selected, $k = 4$, L62, no gate	18.0	15.3	14.0	14.7	10.7	16.7	14.9
Loss-selected, $k = 4$, L62, full loss	21.3	18.7	20.7	9.3	16.7	19.3	17.7
Loss-selected, $k = 4$, L32	48.7	44.7	33.3	31.3	25.3	19.3	33.8
Loss-selected, $k = 4$, L2	62.0	56.7	48.0	35.3	38.0	24.7	44.1
Loss-selected, $k = 4$, L6	65.3	60.7	49.3	48.0	37.3	26.7	47.9
Loss-selected, $k = 8$, L6	68.0	66.7	46.7	52.7	38.0	21.3	48.9
Loss-selected, $k = 16$, L6	70.7	67.3	60.0	56.7	46.0	24.0	54.1
Loss-selected, $k = 32$, L6	68.7	68.0	54.0	55.3	48.0	19.3	52.2
Loss-selected, $k = 64$, L6	70.7	74.0	67.3	50.0	46.7	18.0	54.4
Loss-selected, $k = 4$, L6, $d_{\text{attn}} = 64$	59.3	55.3	43.3	42.0	40.0	22.7	43.8
Loss-selected, $k = 4$, L6, $d_{\text{attn}} = 128$	58.0	56.7	44.0	35.3	37.3	22.0	42.2
Loss-selected, $k = 4$, L6, route gate	52.7	55.3	36.0	38.0	35.3	18.0	39.2
Loss-selected, $k = 4$, L16	64.7	53.3	42.0	33.3	34.7	24.7	42.1
Uniform interval, $k = 4$, L2	63.3	54.7	42.0	37.3	34.0	21.3	42.1
Uniform interval, $k = 4$, L62	21.3	20.0	13.3	9.3	14.0	13.3	15.2
Random, $k = 4$, L62	20.0	14.7	19.3	19.3	16.0	18.0	17.9
Regex/entity, L62	22.0	19.3	14.7	18.0	16.7	20.0	18.4